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SUPERSEDING
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10 May 1951

MILITARY SPECIFICATION

ZIRCONYL NITRATE, DIHYDRATE, ANALYZED REAGENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of zirconyl nitrate, dihydrate.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals
PPP-C-2020	-	Chemicals, Liquid, Dry, and Paste: Packaging of
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1168 - Ammunition Lot Numbering

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Physical characteristics. The zirconyl nitrate, dihydrate, when tested as specified in 4.2.4.1, shall be a white, crystalline, free-flowing powder, free from foreign contaminants.

3.2 Chemical characteristics. The zirconyl nitrate, dihydrate shall conform to the chemical characteristics of table I when tested as specified therein.

TABLE I. Chemical characteristics

Characteristic	Percent by weight	Test paragraph
Assay, as ZrO_2, minimum	45	4.2.4.2
Maximum impurities:		
Solubility	To pass test	4.2.4.3
Chloride (Cl)	0.001	4.2.4.4
Sulfate (SO_4)	0.7	4.2.4.5
Alkalies and alkaline earths	0.10	4.2.4.6
Heavy metals (as Pb)	0.010	4.2.4.7
Iron (Fe)	0.020	4.2.4.8

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the zirconyl nitrate, dihydrate produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 15 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate and shipping containers shall be examined for the following defects using an AQL of 1.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Fiberboard pads or partitions missing or not as specified (where required)
(f) Marking incorrect, missing, or illegible
(g) Unitization not as specified

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4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Physical characteristics. Visually examine the specimen for form, color, flowing characteristics, and the presence of foreign contaminants.

4.2.4.2 Assay, as zirconium oxide. Weigh to the nearest milligram (mg) 0.5 g of the specimen. Ignite, slowly at first, then gradually increasing the heat until the residue is at constant weight. Calculate the percent by weight zirconium oxide as follows:

$$\text{Percent zirconium oxide} = \frac{100A}{B}$$

where: A = Weight of residue in grams, and
B = Weight of specimen in grams.

4.2.4.3 Solubility. Dissolve 1.000 g of the specimen in 20 milliliters (mL) of water. If the solution is not completely clear or possesses more than a very slight opalescence, the specimen does not pass the solubility test.

4.2.4.4 Chloride. Dissolve 1.000 g of the specimen in 20 mL of water. Add 2 mL of 70-percent nitric acid, filter, and add to the filtrate 1 mL of a 2-percent aqueous solution of silver nitrate. If the turbidity is greater than that in a blank to which 0.01 mg of chloride has been added (0.01 mg of chloride equals 0.1 mL of an aqueous solution containing 0.1649 g of sodium chloride per liter), the specimen does not meet the chloride requirement.

4.2.4.5 Sulfate.

(a) Test solution. Dissolve 2.000 g of the specimen in 75 mL of hot water and pour the solution slowly, with stirring, into a mixture of 5 mL of 28-percent ammonium hydroxide and 20 mL of water. Boil for 15 minutes, cool, add enough water to make 100 mL and filter. Retain the filtrate as Solution A.

(b) Procedure. Neutralize 25 mL of Solution A with 37-percent hydrochloric acid. Add 2 mL of 1N hydrochloric acid and 2 mL of a 12-percent aqueous solution of barium chloride and dilute to 50 mL with water. Prepare a control made with 3.5 mg of sulfate (one mL of an aqueous solution containing 0.1814 g of potassium sulfate per liter equals 0.10 mg of sulfate), 2 mL of 1N hydrochloric acid, and 2 mL of the barium chloride solution in the same final volume as with the specimen. If the turbidity in the specimen is greater than that in the control, the specimen does not meet the sulfate requirement.

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4.2.4.6 Alkalies and alkaline earths. Evaporate 50 mL of Solution A from 4.2.4.5(a) to 10 mL, filter, and add to the filtrate 5 drops of 96-percent sulfuric acid. Evaporate and ignite to constant weight, when weighed to the nearest 0.1 mg. Calculate the percent by weight alkalies and alkaline earths as follows:

$$\text{Percent alkalies and alkaline earths} = 100A$$

where: A = Weight of residue in grams.

4.2.4.7 Heavy metals.

(a) Test solution. Dissolve 40.00 g of ammonium citrate (dibasic) in sufficient water to make 90 mL. Add 2 drops of phenol red solution. Make the solution slightly alkaline with 28-percent ammonium hyroxide, and dilute to 100 mL. To remove any lead, shake the solution with 2 or more 30-mL portions of a dithizone solution (25.0 mg of dithizone per liter of chloroform) until the original color of the dithizone solution remains unchanged.

(b) Procedure. Dissolve 1.000 g of the specimen in 20 mL of water and 1 mL of 70-percent nitric acid. Boil for 5 minutes, cool, and dilute to 100 mL. Transfer 10 mL of the solution into a separator and dilute with 5 mL of water. Add 10 mL of ammonium citrate solution described in (a) and sufficient 28-percent ammonium hydroxide to render the solution slightly alkaline to phenol red. Add 10 mL of a dithizone solution in chloroform (5 mg per liter), shake for 1 minute and allow to separate. A red color in the chloroform layer more intense than that in a blank to which 0.01 mg of lead has been added (one mL of a solution containing 0.01598 g of lead nitrate per liter of water) indicates that the specimen does not meet the heavy metals requirement.

4.2.4.8 Iron.

(a) Test solution. Dissolve 10.00 g of potassium thiocyanate in 10 mL of water. Warm the solution to 25° to 30°C, add sufficient n-butanol to make 100 mL and shake vigorously until clear.

(b) Procedure. To 10 mL of the solution from 4.2.4.7(b), add 0.5 mL of 37-percent hydrochloric acid and 15 mL of butanolic potassium thiocyanate. Shake vigorously for 30 seconds and allow to separate. Prepare a control to which has been added 0.02 mg of iron (two mL of a solution of 0.08635 g of ferric ammonium sulfate ($FeNH_4(SO_4)_2 \cdot 12H_2O$) in 10 mL of 10-percent sulfuric acid diluted with water to 1000 mL). If any red color in the clear butanol layer of the specimen is darker than that of the control, the specimen does not meet the iron requirement.

5. PACKAGING

5.1 Preservation. Zirconyl nitrate shall be preserved level A, B or C as specified (see 6.2), and shall be in accordance with the general requirements of PPP-C-2020 and Department of Transportation (DOT) regulations.

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5.1.1 Level A.

5.1.1.1 Unit packing. A quantity of 29.6 (+0.3 or -0) g of zirconyl nitrate shall be unit packed level A in a screw-cap bottle conforming to group A, class 1, type optional glass, style 2 wide-mouth, grade 1 with screw-cap closure A or R, with outer seal A of PPP-C-186. The bottle shall be securely closed to a torque within the range specified by the bottle manufacturer.

5.1.1.2 Intermediate packing. A quantity of 48 bottles of zirconyl nitrate, unit packed as specified in 5.1.1.1, shall be intermediately packed with closures uppermost in two layers, with each layer having 4 rows of 6 bottles per row, within a close-fitting fiberboard box. The box shall conform to grade W5c or W6c of PPP-B-636. Each bottle shall be placed within a close-fitting cell formed by use of full-bottle-height, half-slotted partitions. A fiberboard pad shall be used to completely separate the top and bottom layers of bottles. The partitions and pad shall be formed from fiberboard conforming to grade W5c or W6c of PPP-F-320. The box shall be closed in accordance with PPP-B-636.

5.1.2 Level B.

5.1.2.1 Unit packing. Zirconyl nitrate shall be unit packed level B in the same manner as is specified for level A protection.

5.1.2.2 Intermediate packing. Zirconyl nitrate shall be intermediately packed level B in the same manner as is specified for level A protection except that the box shall conform to class domestic, variety SW, grade 175 or 200 of PPP-B-636; the partitions and pads shall conform to class domestic, variety SW, grade 175 of PPP-F-320.

5.1.3 Level C. A quantity of 29.6 (+0.3 or -0) g of zirconyl nitrate shall be unit packed level C in a manner to assure specified purity and quantity of the zirconyl nitrate from supply source to the initial destination and for a minimum duration of six months.

5.2 Packing. Zirconyl nitrate shall be packed level A, B or C as specified (see 6.2), and shall be in compliance with DOT regulations.

5.2.1 Level A. Two intermediate packs of zirconyl nitrate shall be packed level A with closure uppermost in a close-fitting box conforming to class 3, style optional with veneer faceboards, for a type 2 average load of PPP-B-585. The box shall be fitted with full-box-size pads, lining all inside faces of the box. The pads shall be formed from fiberboard conforming to grade V3c of PPP-F-320. The box shall be closed as specified in PPP-B-585.

5.2.2 Level B. Two intermediate packs of zirconyl nitrate shall be packed level B in the same manner as specified for level A protection except that the box shall conform to grade V3c of PPP-B-636.

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5.2.3 Level C. Two intermediate packs of zirconyl nitrate shall be packed in a manner to assure protection of the contents from supply source to initial destination and for a minimum duration of six months, acceptance by common carrier, compliance with Uniform Freight Classification Rules, National Motor Freight Classification Rules, and the rules applicable to any other intended mode of transportation.

5.2.4 Unitization. Level A and B packs shall be palletized in accordance with the applicable requirements for load type I of MIL-STD-147, using the type IV, grade B pallet of NN-P-71. Level C packs shall be unitized in a manner to assure safe delivery from supply source to initial destination, acceptance by common carrier, and safe stackability in storage for a minimum duration of six months.

5.3 Marking. Level A, B and C containers and pallet loads shall be marked in accordance with MIL-STD-129 and DOT regulations. Each container shall be marked to show lot or batch number and date of manufacture of the zirconyl nitrate.

6. NOTES

6.1 Intended use. Zirconyl nitrate, dihydrate is intended for use as a chemical reagent.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification,
(b) Level of preservation required (see 5.1),
(c) Level of packing required (see 5.2), and

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.5 Precaution. This material should be stored in a dry place, and containers kept tightly closed.

Custodian:

Army - EA

Review activity:

DLA - GS

Preparing activity:

Army - EA

Project No. 6810-B421

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-Z-11144A	ZIRCONYL NITRATE, DIHYDRATE, ANALYZED REAGENT

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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